

Customer Behavior Analysis

In []: PHASE 1: EDA AND CLEANING

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In [7]: # =====  
# Predictive Analytics Project  
# Phase 1: Customer Behavior Analysis & EDA  
# =====  
  
import pandas as pd  
import numpy as np  
  
# Create synthetic dataset  
data = {  
    "CustomerID": [101, 102, 103, 104, 105],  
    "Age": [25, 34, 45, 29, 40],  
    "Gender": ["Male", "Female", "Male", "Female", "Male"],  
    "Income": [45000, 60000, 75000, 50000, 82000],  
    "Spending": [1200, 2500, 3000, 1800, 4000],  
    "Purchase_Date": ["2026-01-15", "2026-02-10", "2026-03-05", "2026-04-20", "2026-05-15"],  
    "Segment": ["A", "B", "A", "C", "B"]  
}  
  
df = pd.DataFrame(data)  
  
# ✅ Add Churn column (0 = Active, 1 = Churned)  
df['Churn'] = np.random.choice([0, 1], size=len(df), p=[0.7, 0.3])  
  
# Save dataset  
df.to_csv("customer_data.csv", index=False)  
print("✅ Synthetic dataset created with Churn column and saved as customer_data.csv")  
  
# Quick checks  
print(df)  
print(df.info())  
print(df.describe())  
print(df.head())  
print("Shape:", df.shape) # rows, columns  
print("Missing values:\n", df.isnull().sum())
```

✓ Synthetic dataset created with Churn column and saved as customer_data.csv

	CustomerID	Age	Gender	Income	Spending	Purchase_Date	Segment	Churn
0	101	25	Male	45000	1200	2026-01-15	A	0
1	102	34	Female	60000	2500	2026-02-10	B	0
2	103	45	Male	75000	3000	2026-03-05	A	1
3	104	29	Female	50000	1800	2026-04-20	C	0
4	105	40	Male	82000	4000	2026-05-12	B	0

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 5 entries, 0 to 4

Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	5 non-null	int64
1	Age	5 non-null	int64
2	Gender	5 non-null	object
3	Income	5 non-null	int64
4	Spending	5 non-null	int64
5	Purchase_Date	5 non-null	object
6	Segment	5 non-null	object
7	Churn	5 non-null	int64

dtypes: int64(5), object(3)

memory usage: 452.0+ bytes

None

	CustomerID	Age	Income	Spending	Churn
count	5.000000	5.000000	5.000000	5.000000	5.000000
mean	103.000000	34.600000	62400.000000	2500.000000	0.200000
std	1.581139	8.080842	15852.444606	1081.665383	0.447214
min	101.000000	25.000000	45000.000000	1200.000000	0.000000
25%	102.000000	29.000000	50000.000000	1800.000000	0.000000
50%	103.000000	34.000000	60000.000000	2500.000000	0.000000
75%	104.000000	40.000000	75000.000000	3000.000000	0.000000
max	105.000000	45.000000	82000.000000	4000.000000	1.000000

	CustomerID	Age	Gender	Income	Spending	Purchase_Date	Segment	Churn
0	101	25	Male	45000	1200	2026-01-15	A	0
1	102	34	Female	60000	2500	2026-02-10	B	0
2	103	45	Male	75000	3000	2026-03-05	A	1
3	104	29	Female	50000	1800	2026-04-20	C	0
4	105	40	Male	82000	4000	2026-05-12	B	0

Shape: (5, 8)

Missing values:

	CustomerID
0	0
Age	0
Gender	0
Income	0
Spending	0
Purchase_Date	0
Segment	0
Churn	0

dtype: int64

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In [36]: # =====
# Step 2: Data Cleaning Script
# =====

import pandas as pd
import seaborn as sns
```

```
import matplotlib.pyplot as plt

# Load dataset (replace with your file name if different)
df = pd.read_csv("customer_data.csv")

# 1. Check dataset structure
print("Shape:", df.shape)
print("\nMissing values:\n", df.isnull().sum())
print("\nData types:\n", df.dtypes)

# 2. Handle missing values
# Example: fill numeric with median, categorical with mode
for col in df.select_dtypes(include=['float64', 'int64']).columns:
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(include=['object']).columns:
    df[col] = df[col].fillna(df[col].mode()[0])

# 3. Remove duplicates
df = df.drop_duplicates()

# 4. Standardize categorical values (example for Gender)
if 'Gender' in df.columns:
    df['Gender'] = df['Gender'].str.strip().str.lower().map({
        'male': 'Male',
        'female': 'Female'
    })

# 5. Convert date columns
if 'Purchase_Date' in df.columns:
    df['Purchase_Date'] = pd.to_datetime(df['Purchase_Date'], errors='coerce')

# 6. Quick outlier check (visual)
if 'Spending' in df.columns:
    sns.boxplot(x=df['Spending'])
    plt.title("Outlier Check: Spending")
    plt.show()

# 7. Save cleaned dataset
df.to_csv("cleaned_customer_data.csv", index=False)

print("✅ Cleaning complete. Cleaned dataset saved as cleaned_customer_data.csv")
```

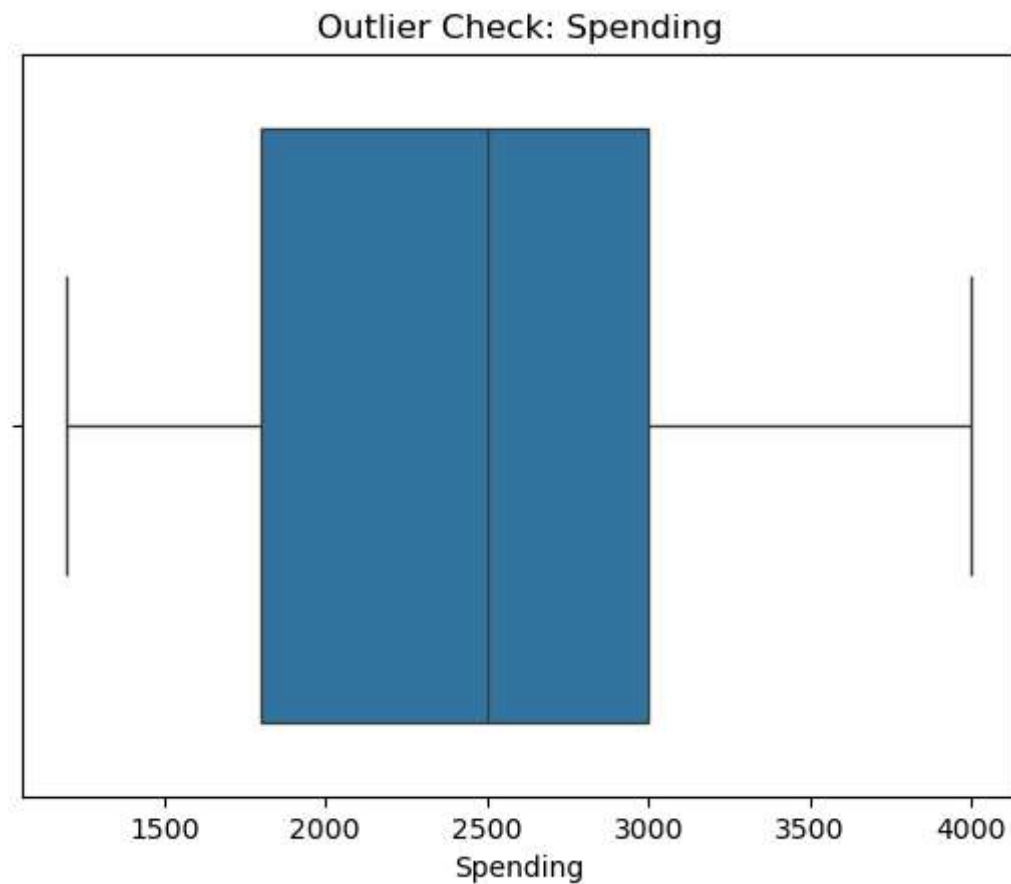
Shape: (5, 7)

Missing values:

CustomerID 0
 Age 0
 Gender 0
 Income 0
 Spending 0
 Purchase_Date 0
 Segment 0
 dtype: int64

Data types:

CustomerID int64
 Age int64
 Gender object
 Income int64
 Spending int64
 Purchase_Date object
 Segment object
 dtype: object



✓ Cleaning complete. Cleaned dataset saved as cleaned_customer_data.csv

```
In [8]: # =====
# Step 3: Univariate Analysis
# =====

import seaborn as sns
import matplotlib.pyplot as plt
```

```
# 1. Age Distribution
sns.histplot(df['Age'], bins=20, kde=True, color='skyblue')
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
# 🖱️ Insight: Shows which age group dominates your customer base.

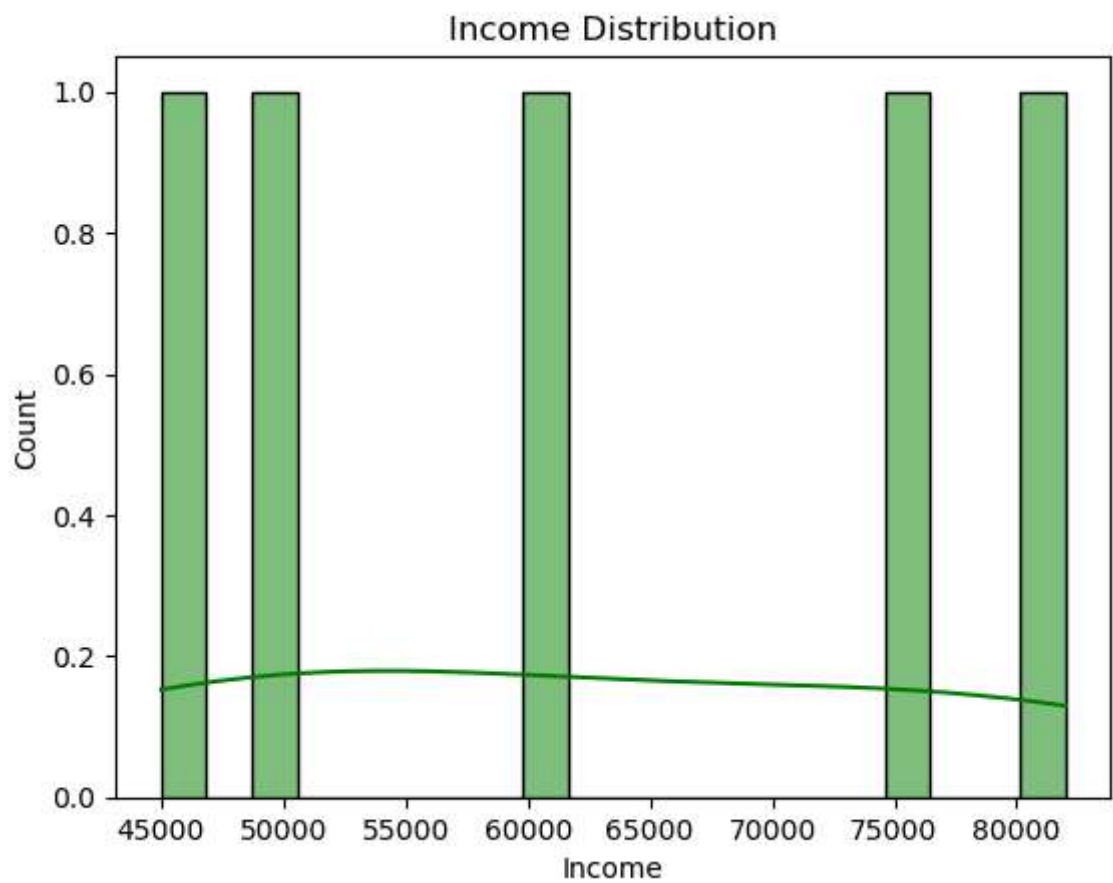
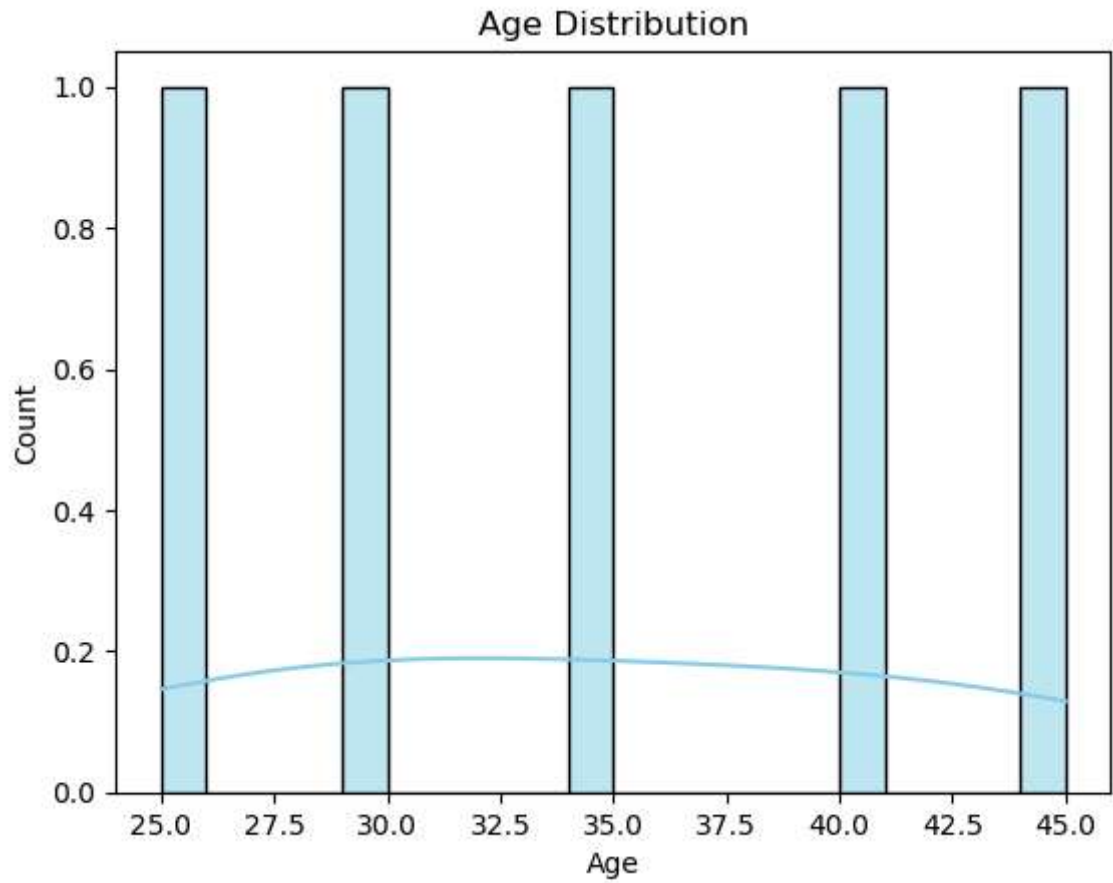
# 2. Income Distribution
sns.histplot(df['Income'], bins=20, kde=True, color='green')
plt.title("Income Distribution")
plt.xlabel("Income")
plt.ylabel("Count")
plt.show()
# 🖱️ Insight: Reveals spread of income levels (low vs high earners).

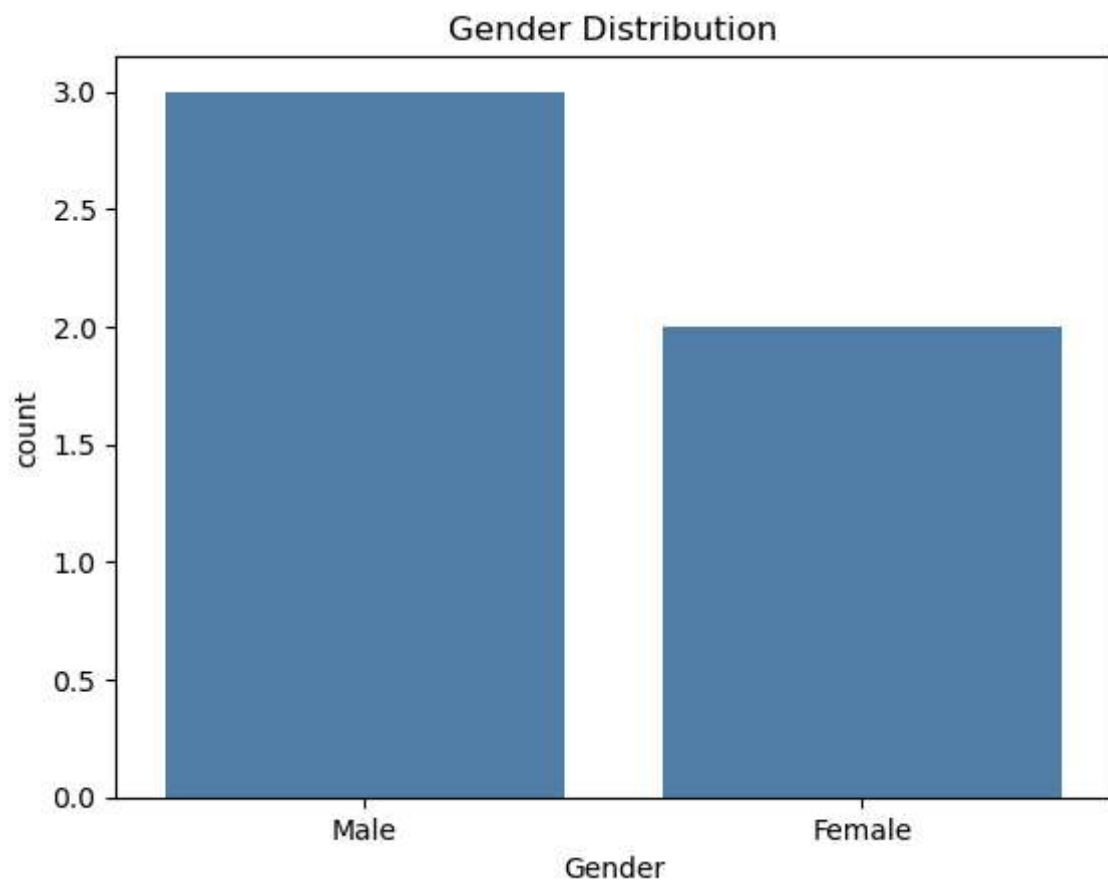
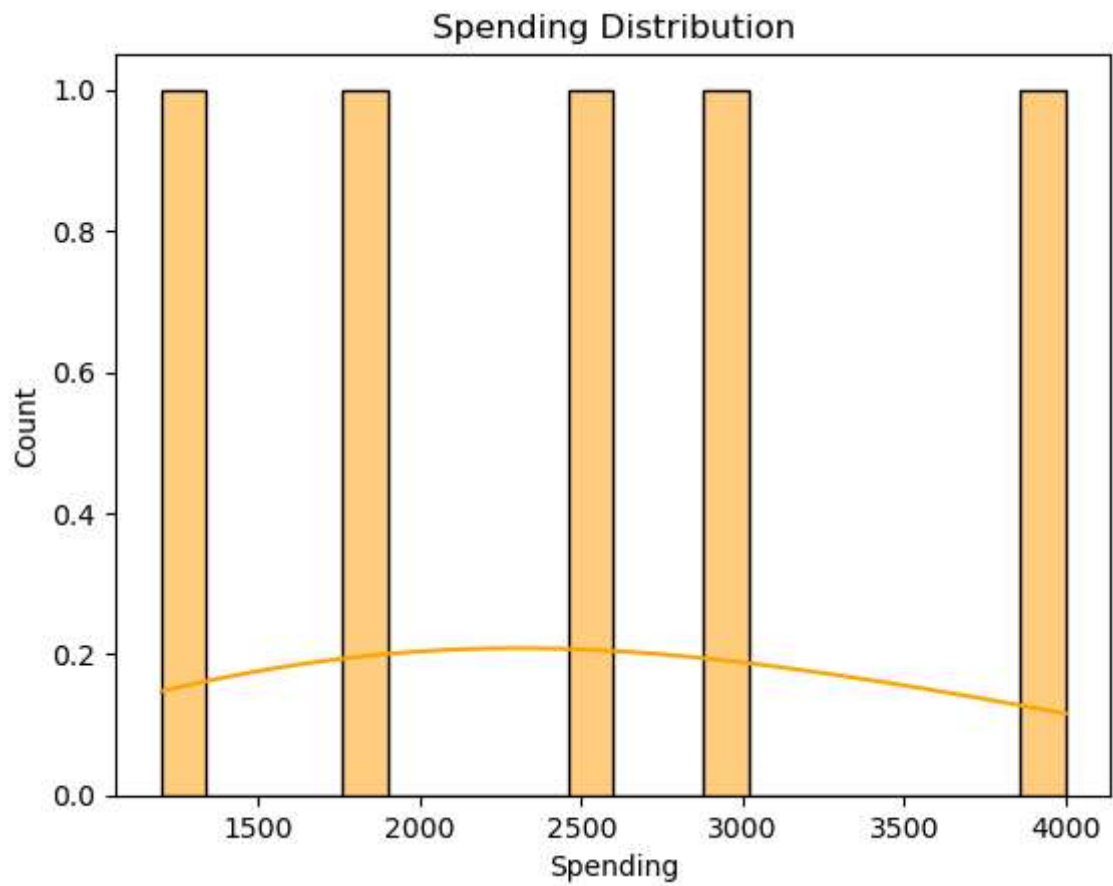
# 3. Spending Distribution
sns.histplot(df['Spending'], bins=20, kde=True, color='orange')
plt.title("Spending Distribution")
plt.xlabel("Spending")
plt.ylabel("Count")
plt.show()
# 🖱️ Insight: Identifies typical spending ranges and big spenders.

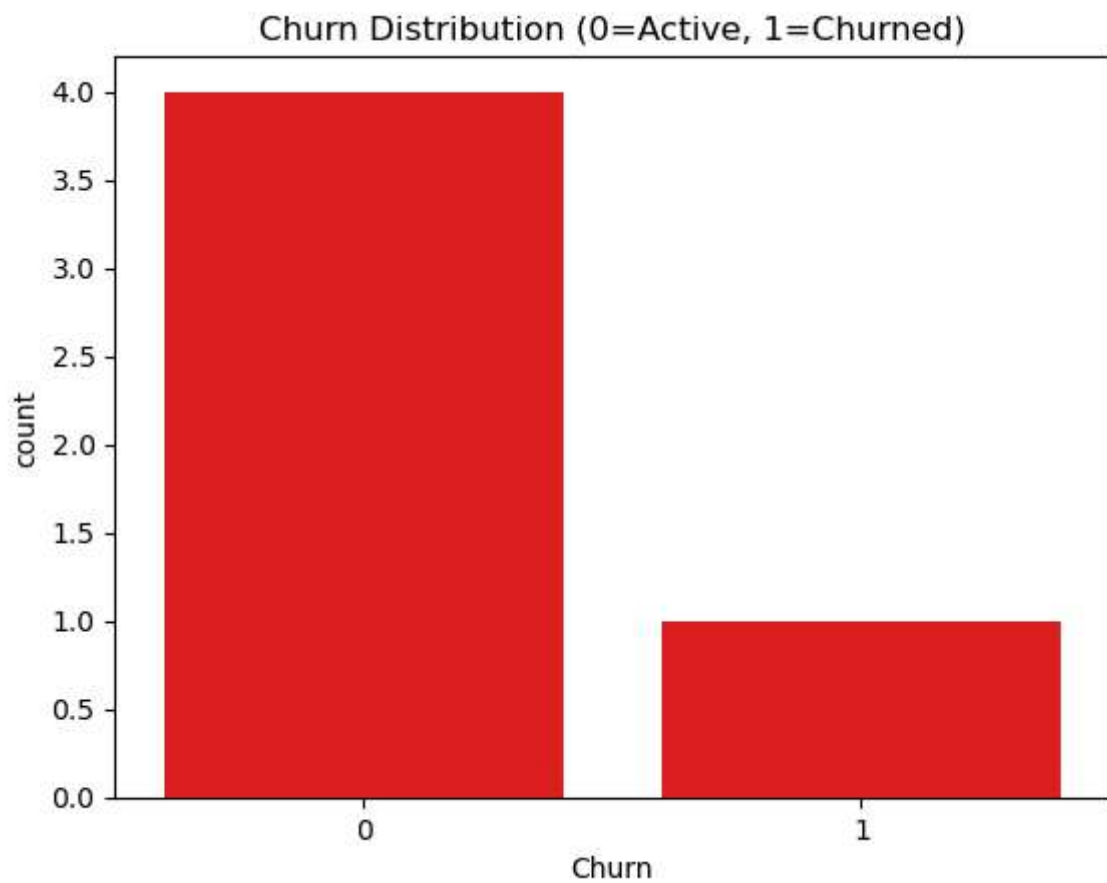
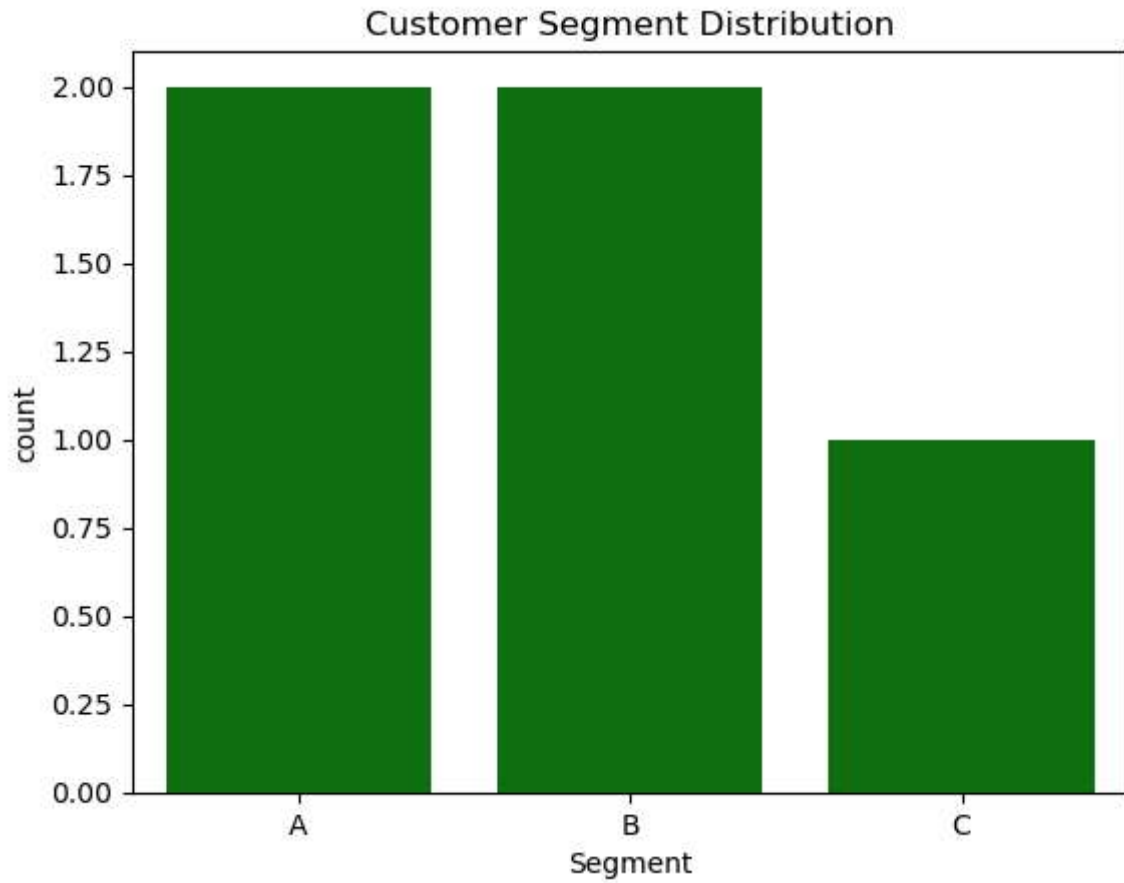
# Gender Distribution
sns.countplot(x='Gender', data=df, color='steelblue')
plt.title("Gender Distribution")
plt.show()

# Segment Distribution
sns.countplot(x='Segment', data=df, color='green')
plt.title("Customer Segment Distribution")
plt.show()

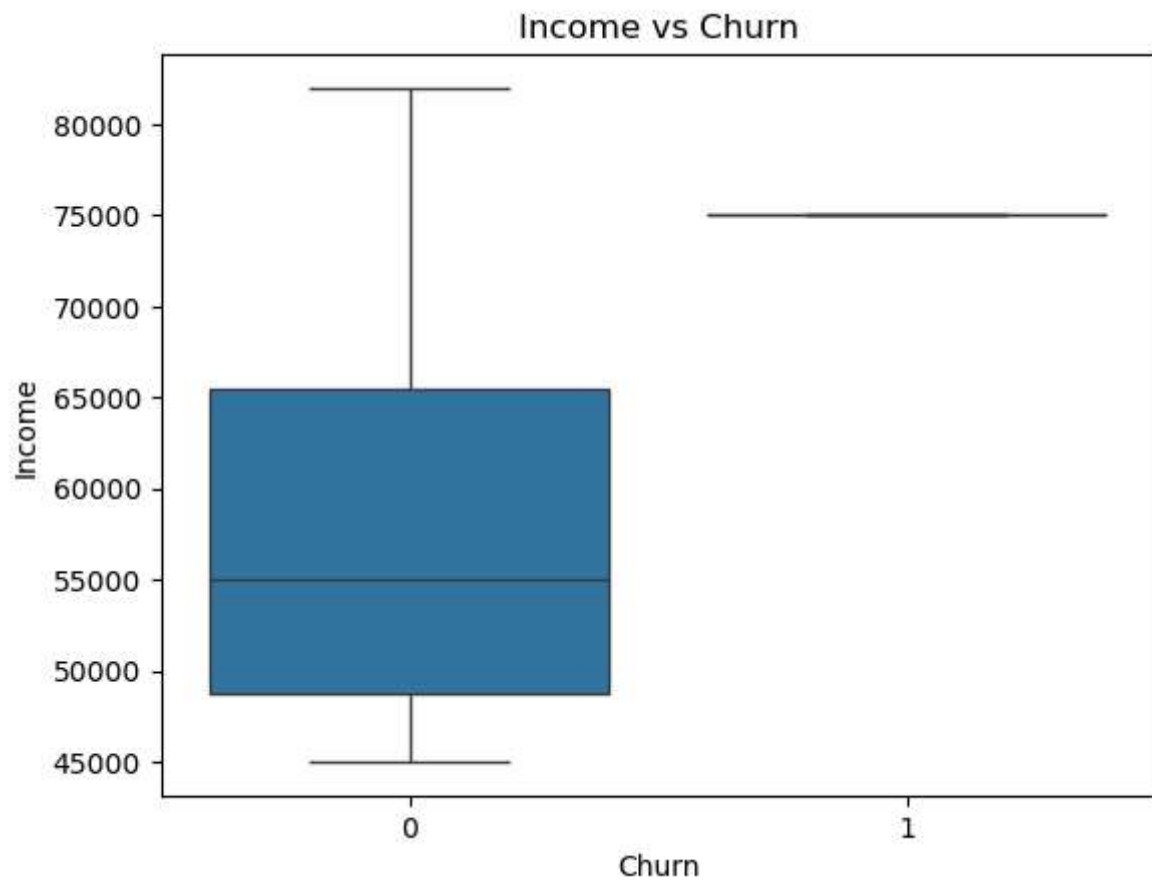
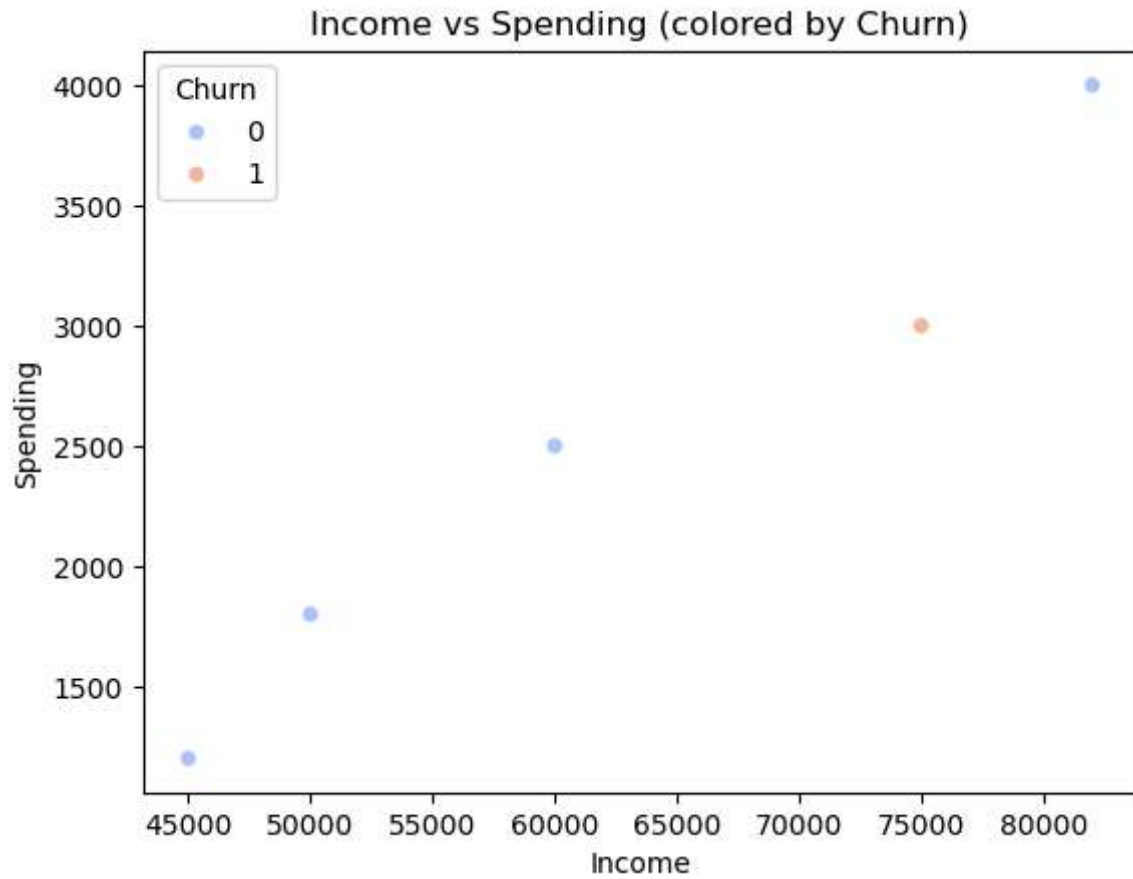
# Churn Distribution (only if column exists)
if 'Churn' in df.columns:
    sns.countplot(x='Churn', data=df, color='red')
    plt.title("Churn Distribution (0=Active, 1=Churned)")
    plt.show()
```

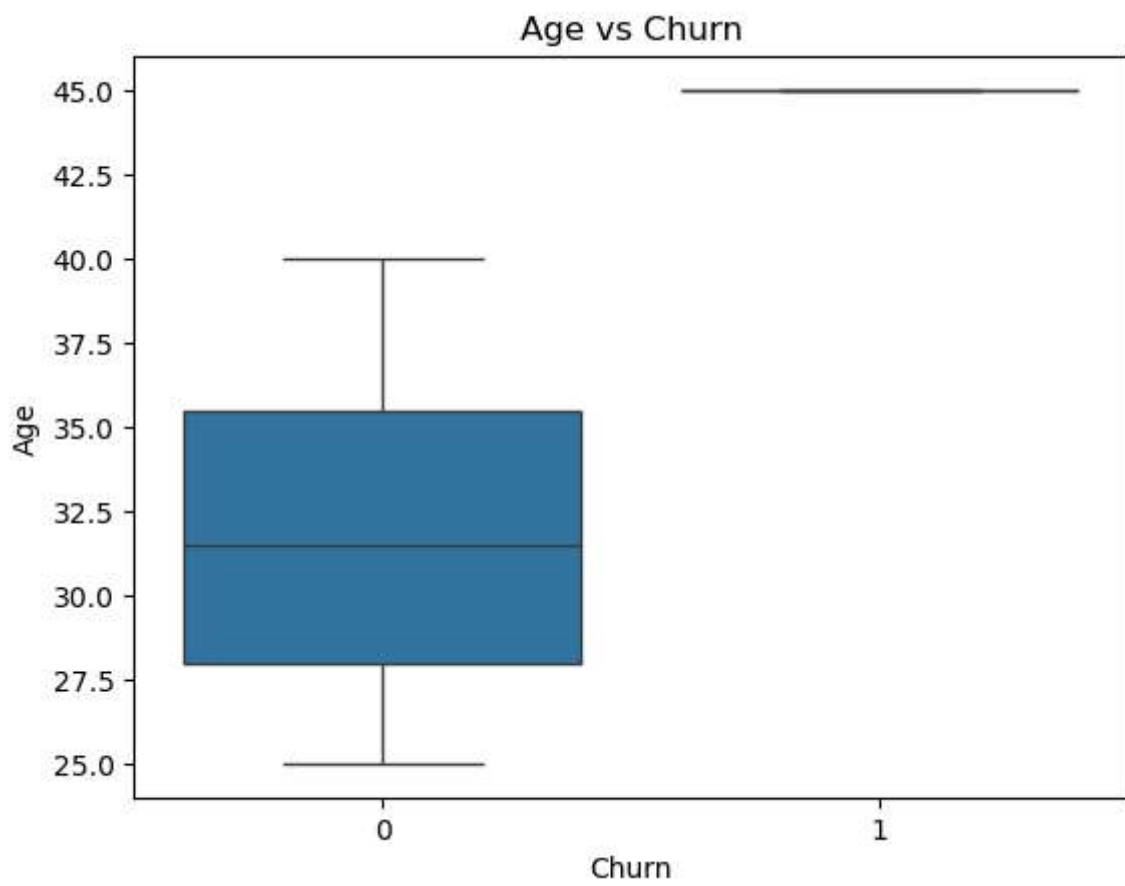
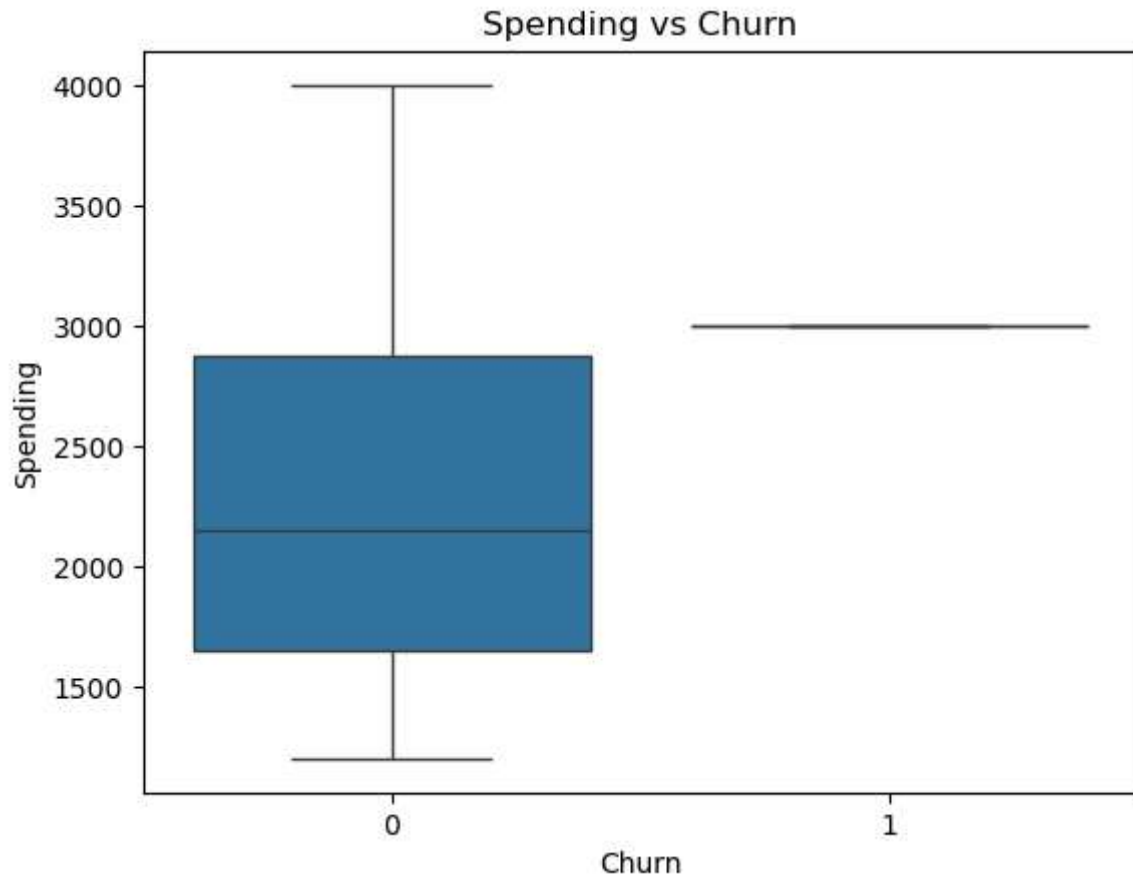


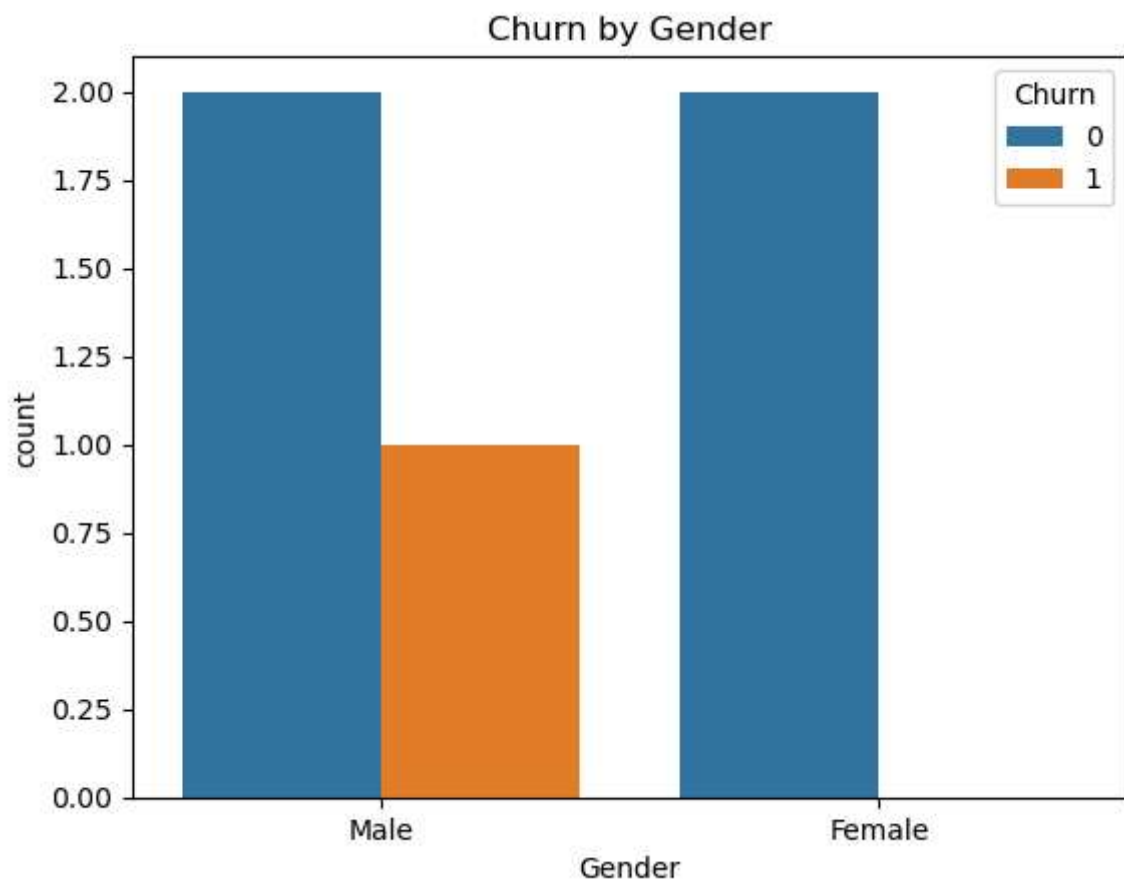
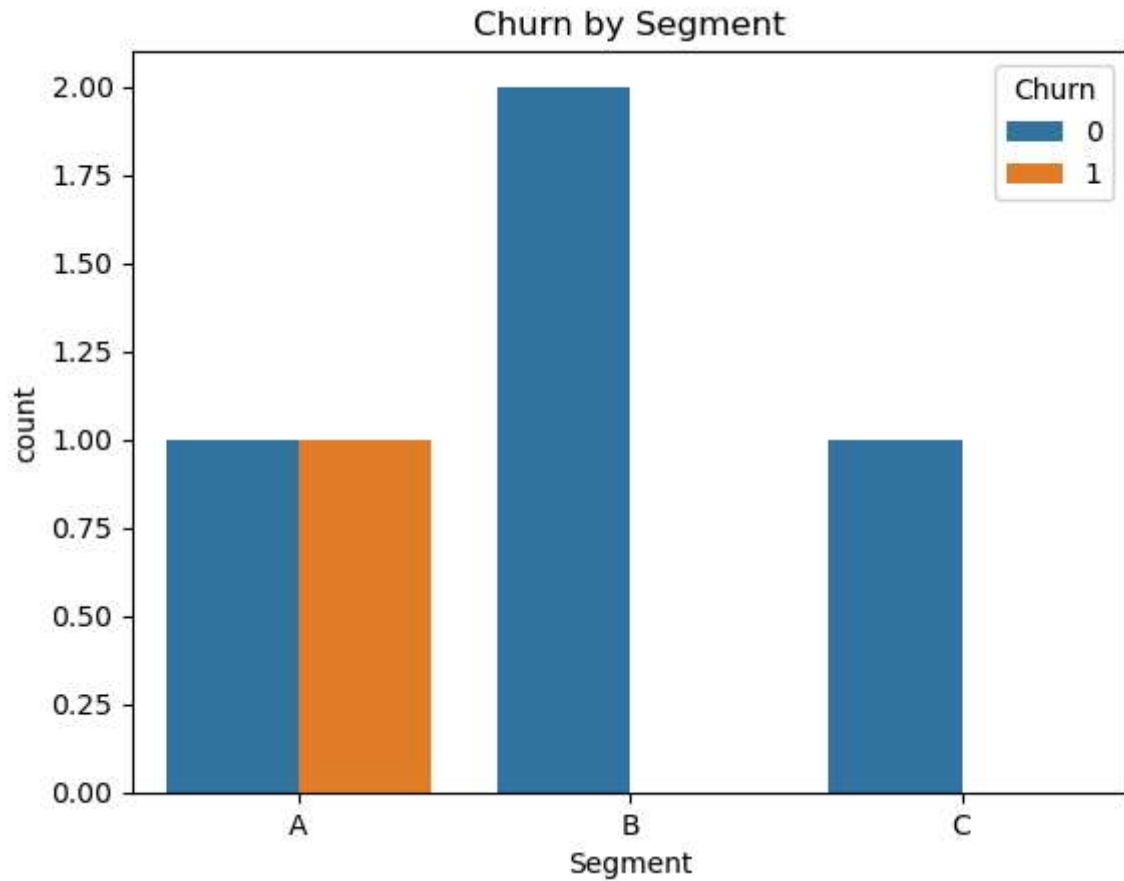





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In [9]: # =====  
# Step 4: Bivariate Analysis  
# =====  
  
import seaborn as sns  
import matplotlib.pyplot as plt  
# Income vs Spending (scatterplot with churn coloring)  
if 'Churn' in df.columns:  
    sns.scatterplot(x='Income', y='Spending', data=df, hue='Churn', palette='coolwa  
plt.title("Income vs Spending (colored by Churn)")  
plt.show()  
  
# Income vs Churn  
if 'Churn' in df.columns:  
    sns.boxplot(x='Churn', y='Income', data=df)  
plt.title("Income vs Churn")  
plt.show()  
  
# Spending vs Churn  
if 'Churn' in df.columns:  
    sns.boxplot(x='Churn', y='Spending', data=df)  
plt.title("Spending vs Churn")  
plt.show()  
  
# Age vs Churn  
if 'Churn' in df.columns:  
    sns.boxplot(x='Churn', y='Age', data=df)  
plt.title("Age vs Churn")  
plt.show()  
  
# Segment vs Churn  
if 'Churn' in df.columns:  
    sns.countplot(x='Segment', hue='Churn', data=df)  
plt.title("Churn by Segment")  
plt.show()  
  
# Gender vs Churn  
if 'Churn' in df.columns:  
    sns.countplot(x='Gender', hue='Churn', data=df)  
plt.title("Churn by Gender")  
plt.show()
```







In []:

